

Mapping the pH heterogeneity of the secretory pathway

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Summary

Introduction: Intracellular pH gradients within the secretory pathway are critical regulators of cellular function, with disruptions linked to diseases such as Congenital Disorder of Glycosylation (CDG) and lysosomal storage disorders. Fluorescence Lifetime Imaging Microscopy (FLIM) uses the biosensor RpHLuorin2 and offers a concentration-independent method for quantitative pH measurement instead of ratiometric measurements. However, extracting accurate pH information from complex FLIM datasets, where signals comprise multiple protonation states and autofluorescence, remains a significant analytical challenge.

Aims: The initial aim of this data analysis project was to develop and critically evaluate an unsupervised guided unmixing FLIM analysis framework for mapping pH of intracellular organelles. However, due to this method's ineffectiveness at providing the results, we decided to move to a supervised approach. The supervised method's accuracy and limitations were tested, and a future integration pathway using next-generation unsupervised algorithm was proposed.

Methodology: Recombinant RpHLuorin2 was calibrated in buffers with known pH values (4.5, 5.5, 6.0, and 7.5) to generate reference fluorescence decay curves. These references were used as fixed components in a Non-negative Matrix Factorisation (NMF) algorithm within the Hyperspectral Image Analysis (HIA) software, implementing a guided analysis framework. This method was applied to cellular FLIM data, and spatial pH maps were generated from component concentration maps through weighted average calculations in ImageJ and MATLAB. Performance was evaluated based on signal-to-noise ratio (SNR), resolution of intermediate pH values, and agreement with expected pH values.

Key Results: The guided unmixing method successfully generated spatial pH maps capable of distinguishing broad pH ranges, with clear colour-coded trends corresponding to increasing pH observed across calibration samples. However, several critical limitations were identified, including poor resolution of some pH values, absolute dependence on spatial binning (4 x 4) to achieve sufficient SNR, and systematic offsets between calculated and nominal pH values. The four-component model demonstrated over-parameterisation, with intermediate references contributing minimally to the final analysis. Background noise from overlapping spectral signatures also contributed to the complexity of interpretation.

Discussion and Conclusion: The results of this data analysis validate the principle of component-guided FLIM unmixing for determining the pH values. They also define the constraints of this study, namely SNR limitations and model rigidity. The findings demonstrate that while supervised approaches can generate qualitative pH maps, they are vulnerable to systematic errors when the reference used as a guide for analysis is not suitable for determining the pH of the organelles under *in-cellulo* conditions. The necessity for spatial binning highlights the photon budget limitations in FLIM experiments. These results demonstrate the necessity for a shift from supervised toward an unsupervised analysis framework. Integrating this proposed process into the uFLIM algorithm and phasor S-FLIM, which uses model-blind computational methods, points towards a future direction for achieving precise, high-resolution, and quantitative mapping of organelle pH. Such integration could enable true multi-parameter cellular physiology and allow for the analysis of quantitative, dynamic readouts of cellular function.

1. Introduction

1.1 Importance of Identifying Intracellular pH Gradients

In eukaryotic cells, a distinct biochemical environment is present due to internal compartmentalisation (Gabaldón and Pittis 2015), which in turn regulates organelle function through differences in luminal pH across each compartment (Jain and Zoncu 2022). The secretory pathway provides an ideal model for observing pH heterogeneity, as it exhibits a finely tuned pH gradient. A near neutral pH of approximately 7 is observed in the endoplasmic reticulum (ER), a moderately acidic pH of approximately 6.0 - 6.7 is observed in the Golgi apparatus, while the secretory vesicles (pH approximately 5.2) and lysosomes (pH approximately 4.7) demonstrate a more acidic environment (Linders et al. 2022) as seen in figure 1.

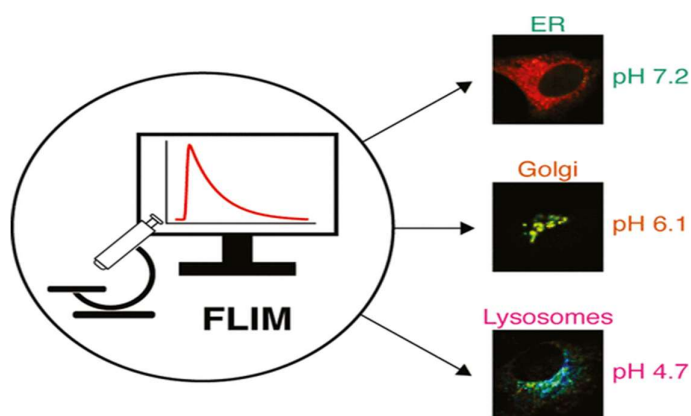


Figure 1. The differences in pH of three different organelles observed after FLIM quantitative analysis (Linders et al. 2022).

This pH gradient varies between organelles and is essential for controlling critical cellular processes. It actively supervises the folding and stability of nascent proteins, modulates specific luminal enzyme activity, triggers conformational changes in trafficking receptors such as ERGIC-53 and KDEL, and ensures that cargo is packaged and transported according to regulatory requirements (Kamiya et al. 2007) (Linders et al. 2022). Disruptions to pH homeostasis have been found to be directly linked to the development of diseases, like Congenital Disorders of Glycosylation (CDG) due to pH imbalance in Golgi, and various lysosomal storage diseases (Linders et al. 2020). Therefore, developing methods for accurately measuring pH with high spatiotemporal accuracy at the subcellular level is of

great importance and key to furthering our understanding of cell biology and the mechanisms of certain diseases (Zhang et al. 2024) (Linders et al. 2022).

1.2 Evolution of pH Biosensors

The surge in interest regarding the effects of drugs and mutations on cellular pH levels had led to the rapid evolution of fluorescent biosensor technology, which utilises pH dependent fluorescent lifetime imaging microscopy (pHFLIM) alongside deep learning (Rennick et al. 2022). The early genetically encoded sensors used pH sensitive GFP (pHLuorin) for imaging (Royle et al. 2008), which captured data based on intensity, making it susceptible to false positives due to different expression levels, organelle movement, photo bleaching, and excitation intensity variations (Martinez et al. 2012). To alleviate such errors, ratiometric pHLuorin (RpHLuorin) was developed, which improved readability by calculating the ratio of photon emission intensities from two different excitation wavelengths (Mahon 2011). This helped normalise image capture for sensor concentration.

While this ratiometric method enabled sequential image capture, it is a slow process that could increase the chances of encountering misalignment errors in highly dynamic organelles and could also result in high variations in ratio values due to its high sensitivity to background fluorescence (Linders et al. 2022). Moreover, when attempting to measure the sensor signal in acidic environments, the amount of noise captured increases, thus leading to a lower signal-to-noise ratio (SNR) (Liu et al. 2006). This can yield inaccurate pH values in such areas, especially in dim compartments. Figure 2 illustrates the differences in pH calculation between the bio-sensing technologies.

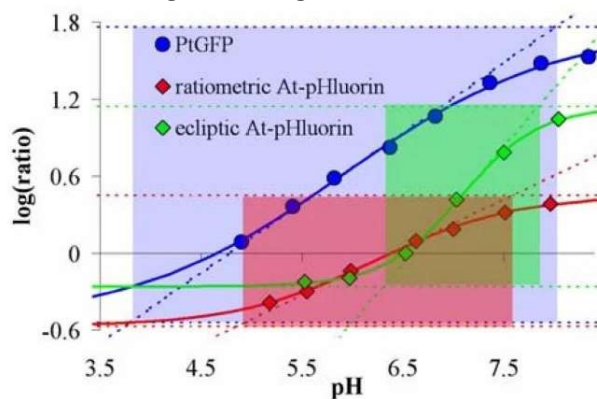


Figure 2. Comparison of three different pH probe results on a log-log scale. The coloured rectangles show areas of best responsiveness. RpHLuorin seen in red has better pH value capture (Schulte et al. 2006).

1.3 Using FLIM for Quantitative Analysis

Fluorescence Lifetime Imaging Microscopy (FLIM) is a microscopy technique capable of generating a highly precise spatial map of fluorescence lifetime at the individual pixel level (Torrado et al. 2024). The electrons in the molecule absorb light which excites them to a higher energy level. When these excited electrons subsequently return to their ground state, light is emitted and fluorescence occurs. FLIM utilises ratiometric pHluorin2 (RpHluorin2) and is capable of overcoming the limitations of previous biosensors (Linders et al. 2022). This is achieved by measuring the fluorescence lifetime (τ), which is the average time a fluorophore spends in the excited state before photon emission (Datta et al. 2020).

A critical advantage of FLIM is that τ is an intrinsic molecular property (Berezin and Achilefu 2010), consequently, changes in fluorophore concentration, excitation light intensity, and photo bleaching do not affect the value (Jain et al. 2009). Such stability makes it particularly useful for measuring the local microenvironment parameters such as pH, ion concentration, and molecular interactions. Furthermore, FLIM is capable of capturing quantitative pH measurements in a single excitation wavelength scan using the pH dependent lifetime shift found in RpHluorin2 (Linders et al. 2022). Therefore, it is also effective for capturing rapid cellular dynamics without errors or distortions due to organelle motion (Bower et al. 2018). The effectiveness of FLIM with RpHluorin2 was successfully demonstrated by Linders et al. (2022), where the pH of the entire secretory pathway was mapped and it was identified that the pH lifetime range of FLIM was approximately 4.5 to 7.5 (Refer to figure 3).

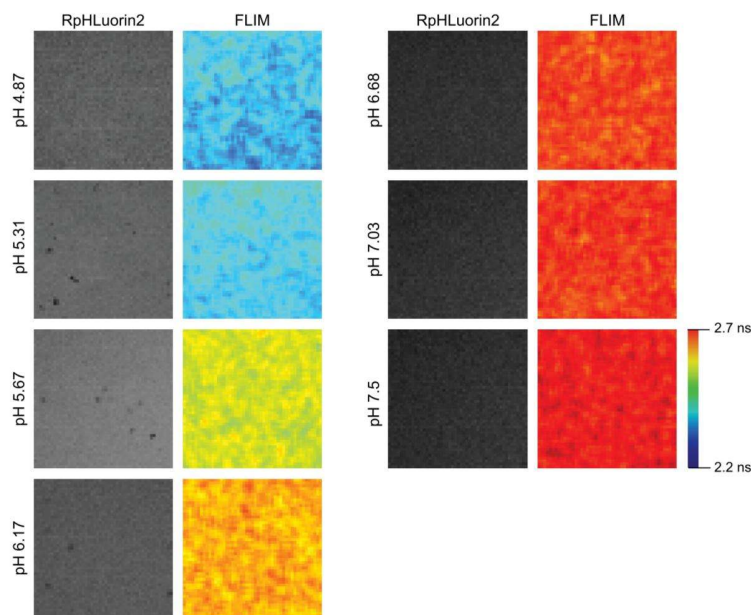


Figure 3. FLIM of recombinant RpHLuorin2 showing pH values based on image intensity (Linders et al. 2022)

1.4 Data Analysis of the pH Heterogeneity of the Secretory Pathway

In recent years, hyperspectral imaging has gained popularity due to its effectiveness when modified for analysis and the growing interest in research on electronic imaging (Khan et al. 2018). This has been particularly evident in pharmaceutical research, for applications such as spectroscopy and chemical imaging (Gowen and Amigo 2013), food sciences, for chemometrics to analyse food structure, composition and quality (Amigo et al. 2013), and in biochemistry and biomedicine, for studies of biological samples to demonstrate the feasibility of certain drugs on organisms (Zhang et al. 2014). The HIA (Hyperspectral Image Analysis) program was developed to analyse the hyperspectral images, especially for quantitative chemical imaging and analysis of Raman data (Masia et al. 2013).

While FLIM has been successfully used to determine the pH of the secretory pathway, it has not yet been optimally employed for data analysis in a cellular context, as the fluorescence decay at each pixel does not follow a single rate constant (Wüstner 2022). The data collected represents a linear combination of multiple components, including those collected from the pH sensor in various protonation states, cellular fluorophores (such as those from NADH and FAD), and other co-expressed fluorophores. If the decay curves are fitted to standard mono-

exponential or bi-exponential models, which are computationally intensive, the outcome would result in an often-incorrect functional form that could struggle with areas with low photon counts (Warren et al. 2013) (Datta et al. 2020).

To overcome such shortcomings in data analysis, several advanced computational methods have been developed. Phasor analysis uses a fit-free, graphical approach to transform the lifetime decay at each pixel into a singular point on a 2D phasor plot (Guo et al. 2025).

Although this method can quickly identify distinct components, it lacks quantitative precision for resolving closely spaced lifetimes (Soltani et al. 2025). Another method is supervised unmixing or pattern matching, which decomposes data into a linear combination of pre-defined reference decay patterns obtained from control samples (Gregor and Patting 2014). While this method can provide accurate analysis, it may lead to inaccuracies if the reference sample dynamics do not perfectly match the samples being experimented on, due to environmental differences between *in vitro* and *in cellulo* conditions.

The most advanced method, unsupervised unmixing FLIM (uFLIM), employs the Non-negative Matrix Factorisation (NMF) algorithm to separate a FLIM dataset into its constituent spatial distributions and temporal dynamics without any prior knowledge of the component decay profiles (Masia et al. 2022). NMF is a dimensionally reduction technique that factorises the data matrix D (pixels x time channels) into the product of a spatial concentration matrix S (pixels x components) and a temporal dynamics matrix T (components x time channels). The non-negative constraint ensures that all the results calculated are physically interpretable where the values in both matrices are greater than or equal to zero. uFLIM can be used to visualise different fluorescent components separately, as well as to reveal interaction parameters such as FRET rates, offering a powerful method for analysing complex biological samples. The process is illustrated in figure 4.

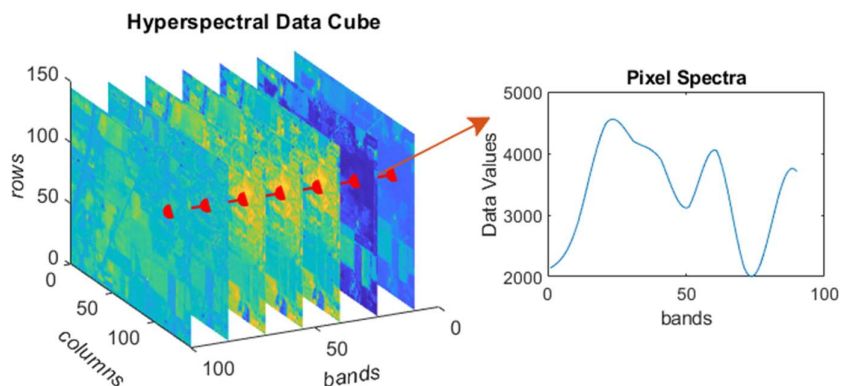


Figure 4. Hyperspectral imaging data cube used to acquire hyperspectral data of varying intensities as pixels which is then plotted as the pixel spectrum vector (MATLAB 2026).

1.5 Project Aims and Hypothesis

The main aim of this data analysis project was to propose a method for bridging the gap between the currently available sensor technology and the limitations of computational analysis capabilities. A guided unmixing FLIM analysis was implemented to evaluate the intracellular pH mapping. We hypothesised that the decay curves obtained from *in vitro* RpHLuorin2 could be used as effective references for guided analysis when comparing to the probe cellular FLIM data, enabling the generation of accurate pH maps.

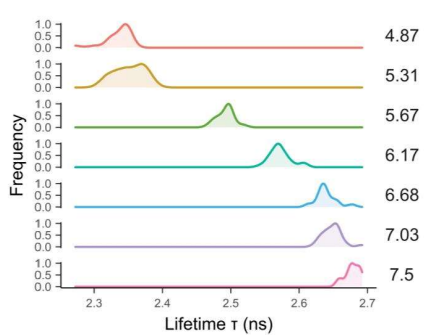


Figure 5. Average lifetime histograms of recombinant RpHLuorin2 used to identify key pH values to use as a guide for analysis (Linders et al. 2022). The distribution lifetimes in the *in vitro*, using the fitting method, is an indication that the method could have some limitations due to low SNR.

To obtain the dataset required for guided analysis, the pH dependent in vitro fluorescence lifetime decay of recombinant RpHLuorin2 was calibrated at specific key pH values that exhibited good peak separation (4.5, 5.5, 6.0, and 7.5), as illustrated figure 5. These peaks correspond to the different pH dependent lifetime states directly corresponding to environmental pH changes (Linders et al. 2022). Selecting only the pH values with good peak separation ensures the accuracy of pH measurements. Including pH values with overlapping peaks could introduce noise and crosstalk, leading to inaccurate analysis.

Using these calibrated pH values as fixed spectral profiles, they were implemented into a guided NMF algorithm to analyse the FLIM data from probe cellular samples. In guided NMF, the temporal dynamic matrix T is fixed using the pre-generated reference decay curves, and the algorithm solves only for the spatial concentration matrix S . This helps determine how much of each reference pH component contributes to every pixel in the image.

After assessing the accuracy of pH values against expected values, alongside the spatial resolution and sensitivity of the FLIM images, spatial pH maps are generated. By comparing the findings from our analysis method to the wider existing literature, we can critically analyse the methodological limitations. This facilitates the proposal and justification for formulating a new integrated framework for quantitative pH mapping that combines the strengths of this experimental project with future innovations to achieve clearer pH value results.

2. Experimental Procedures

2.1 Description of the FLIM Dataset

The recombinant pH data used in this analysis was taken from Linders et al. (2022). The process used by Linders et al. (2022) to acquire this data is as following. The recombinant RpHLuorin2 protein was suspended in *E. coli* and purified using nickel-nitrilotriacetic acid (Ni-NTA) affinity chromatography. This purified protein was diluted and separated into different buffers for calibration at the predefined pH values of 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5. These buffers were composed of a mixture of 125 mM KCl, 25 mM NaCl, and 25 mM of MES, if $\text{pH} \leq 6.5$, or HEPES, if $\text{pH} \geq 7.0$. These buffers were then equilibrated to the required temperature, with the final pH of 7.5 adjusted to 37°C.

FLIM data were acquired from each mixture using a time-correlated single-photon counting (TCSPC) system equipped with a pulsed white light laser at an 80 MHz repetition rate and a HyD detector. In this case, a Leica SP8 SMD was used. The sensor was excited at 488 nm, and emission was collected at a higher wavelength between 502 and 530 nm. For each pH buffer, a global fluorescence decay curve was generated which was done by averaging the number of photons from the entire field of view. This creates a single, high SNR decay trace that accurately reflects the pH value under the specified buffer conditions.

It should be noted that the original author, Linders et al. (2022), did not provide pixel size information in their paper. As a result, it was not possible to display scale bars with physical dimensions in the figures presented in this analysis. However, the absence of this information does not affect the analysis performed, as the methods employed operate on relative intensities and decay kinetics independent of absolute spatial scaling.

2.2 Data Pre-Processing and Optimisation

Before we began the analysis process, several pre-processing steps were implementing by testing and determining the best settings. This was done to ensure data quality and to optimise the performance of the factorisation algorithm. Initially, examination of the full decay curves revealed an undesirable secondary artefact peak at approximately 8 ns after the initial excitation peak (Illustrated in figure 6). This peak was determined to be linked to instrumental errors rather than genuine pH dependent fluorescence decay. To ensure that

only the relevant sensor decay was analysed and to maintain data accuracy, the spectral range was limited to 6 ns for all subsequent analyses. This cropping removes the error while preserving the complete region of the decay (0 to 6 ns) which helps determine the pH accurately.

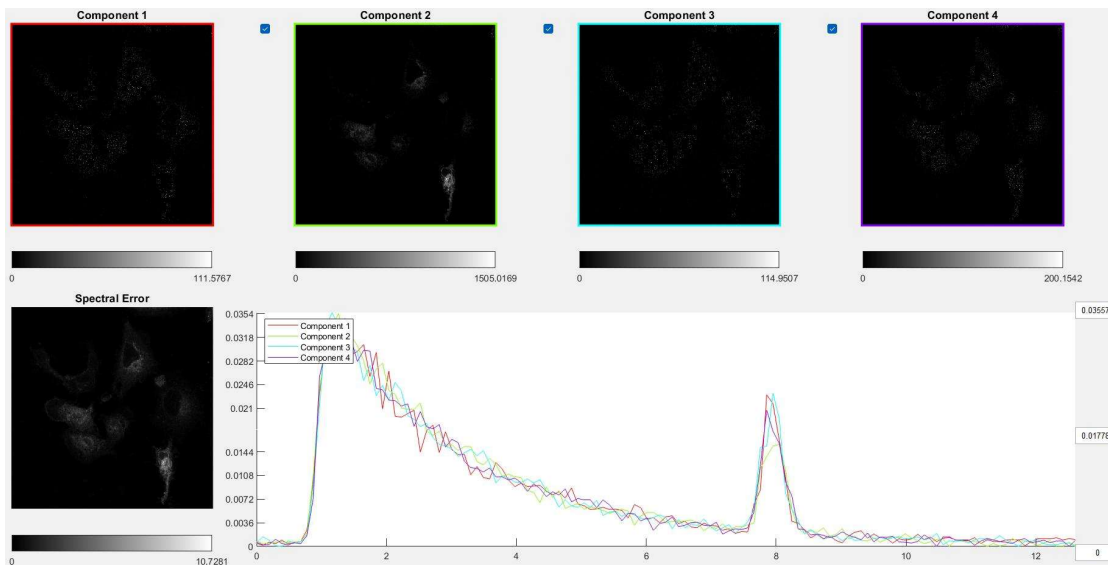


Figure 6. The resulting analysis result of the pH 4.5 probe dataset without limiting spectral range and without binning. A second peak can be seen at approximately 7 ns.

Signal to noise ratio (SNR) optimisation was performed through spatial binning. SNR is a metric that compares the power of the signal to the background noise, with a higher SNR indicating a clearer and stronger resultant image as even a small fluctuation in pH can be accurately detected, while a lower ratio means the data is obscured by noise (Semmlow, 2018). Spatial mitigates low photon counts by averaging adjacent pixels, thus reducing them. Various binning sizes, ranging from 1 x 1 to 128 x 128, were tested. A binning of 4 x 4 was found to be optimal, as it provided a significant increase in SNR per image element and the most accurate pH reading without suffering from excessive loss of spatial resolution. The effect of different binning factors on image quality is presented in figure 7.

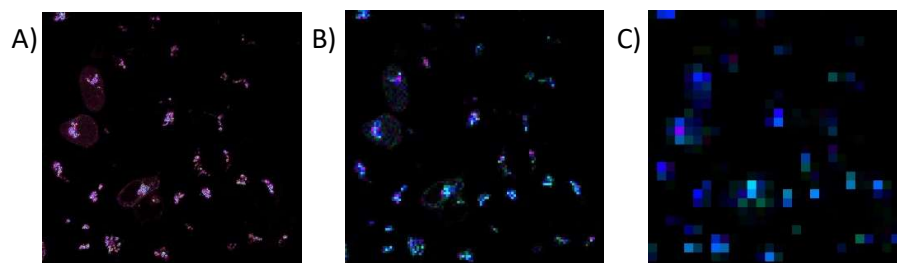


Figure 7. Images showing the coloured map of pH6.5 probe dataset found with different binning sizes. A) 1x1 binning does not show good SNR B) 4x4 binning shows good SNR without suffering from resolution loss C) 16x16 binning starts to suffer from loss of resolution

Partial noise whitening was applied to improve factorisation stability. FLIM data is acquired by photon counting, which means that bright signals, found at the initial peak of the decay curve, have higher absolute noise variance, while dim signals, seen at the tail end of the decay, have lower variance. This means that noise is not consistent across the dataset, and bright pixels contribute more to factorisation error due to their larger values, even if their relative error is small. Partial noise whitening, a pre-processing step, scales the data to achieve uniform variance across the dataset before factorisation. By whitening the data first, the noise variance becomes more uniform, preventing the algorithm from being overwhelmed by the high noise variance in the early parts of the decay curve. More importantly, it allows the algorithm to properly analyse the low signal region of the decay which encodes subtle lifetime differences. The effect of partial noise whitening on data quality is illustrated in figure 8.

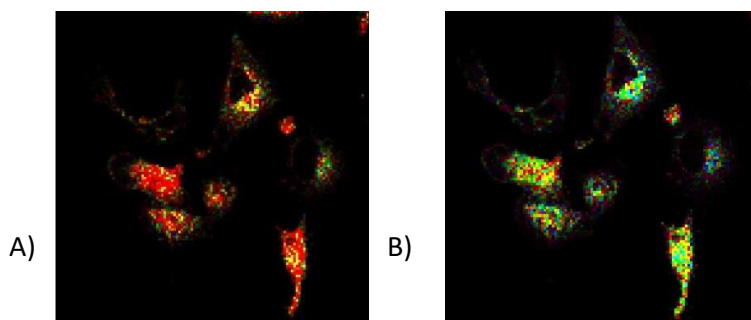


Figure 8. These images are the analysis results with and without partial noise whitening. A) Partial noise whitening used B) Partial noise whitening not used

2.3 Generation of Reference Data for Guided Analysis

After the pre-processing step, reference decay curves were generated from the recombinant calibration data to serve as fixed components for guided non-negative matrix factorisation (NMF) analysis.

The four pH values (4.5, 5.5, 6.0, and 7.5) exhibiting the greatest spectral variations were selected from the completed calibration set and imported into the Hyperspectral Image Analysis (HIA) software (Masia et al. 2022). For each of these four pH datasets, the spectral average decay curves were extracted and normalised to have equal integrated area. This normalisation ensures that each reference curve represents a true probability density function for photon emission at that specific pH, independent of absolute intensity variations.

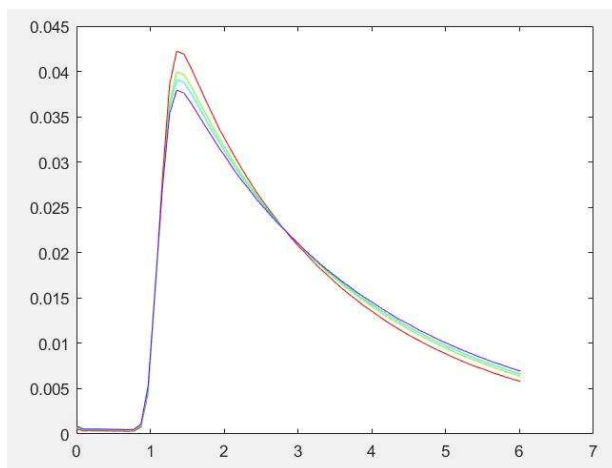


Figure 9. The spectra used as the fixed reference matrix point for guided analysis obtained

In a new workbook, the calculated normalised curve data were compiled to obtain four probability density functions representing the normalised spectral fingerprint for each reference pH. These four curves (refer to figure 9) were assembled into a single ASCII text file to act as the fixed reference matrix for all subsequent guided analyses.

2.4 Non-Negative Matrix Factorisation (NMF) Framework

NMF is an algorithm forming the core of the analytical approach used in this study. It works by dimensionally reducing and separating the source image. The fundamental principle of NMF can be expressed using the expression: $D \approx S \times T$, where D is the original data matrix (pixels x time channels), S is the spatial concentration matrix (pixels x components), and T is the temporal dynamics matrix (components x time channels).

For FLIM data analysis, the decay curve measured at each pixel can be interpreted as a linear combination of a small number of component decay curves. Each component has a specific decay profile (stored in T) and a spatial distribution indicating how much of that component is present in each pixel (stored in S).

For good physical interpretability, non-negative constraint is important. Unlike other factorisation methods, like principal component analysis which can yield negative values, NMF forces all values in both S and T to be greater than or equal to zero. This need for a positive value works in accordance to the fact that in fluorescence measurements, photons are used which cannot be negative, and a component obtained from a pixel's signal cannot have a negative value. The constraint ensures that the decomposition gives interpretable results.

In our analysis, a guided NMF approach was used. Instead of using an unsupervised approach which allows the algorithm to calculate the value of S and T from the data, the value of T matrix was fixed using the pre-generated reference decay curves corresponding to pH 4.5, 5.5, 6.0, and 7.5. The algorithm was used to only determine the spatial concentration matrix (S) which can help determine how much of each reference pH component contributes to each pixel in the image.

2.5 Guided NMF Analysis of Calibration and Cellular Data

Using the determined framework, guided NMF analysis was performed on both the recombinant and probe datasets. In the HIA software, a cellular FLIM dataset of recombinant protein with a known pH value was selected and loaded. The spectral range was limited to 6 ns, and the image was binned for x- and y- axis set to 4. This data was then processed using FSC3 (Factorisation into Susceptibilities and Concentrations of Chemical

Components) with the number of components set to 4, and the minimum and maximum iterations set to 1. FSC3 is a critical component of the HIA software used to analysis hyperspectral Coherent anti-Stokes Raman Scattering (CARS) data to generate concentration maps of chemical components which is based on the NMF analysis method (Boorman 2019). The number of iterations, in this case, are irrelevant as the software is performing a projection where it searches for the abundance map of the components as defined by the supplied decay curves acting as the guided analysis reference matrix, only analysing the data for its spatial concentrations. The single method was selected for this analysis, as the knock-out method is typically used for calculations where the guide is not provided. Partial noise whitening and guided analysis was selected, and when prompted, the ASCII text file containing the four reference decay curves was selected.

The output from FSC3 processing was a stack of the four images with pH decay graphs corresponding to the spatial concentration maps for each reference pH component. By changing the normalisation setting from Concentration to Spectrum Integral, the dominant component and the differences between each were observed and noted. Figure 10 shows the results of analysing pH 5.0 and 6.5 recombinant datasets. Each resultant output was saved.

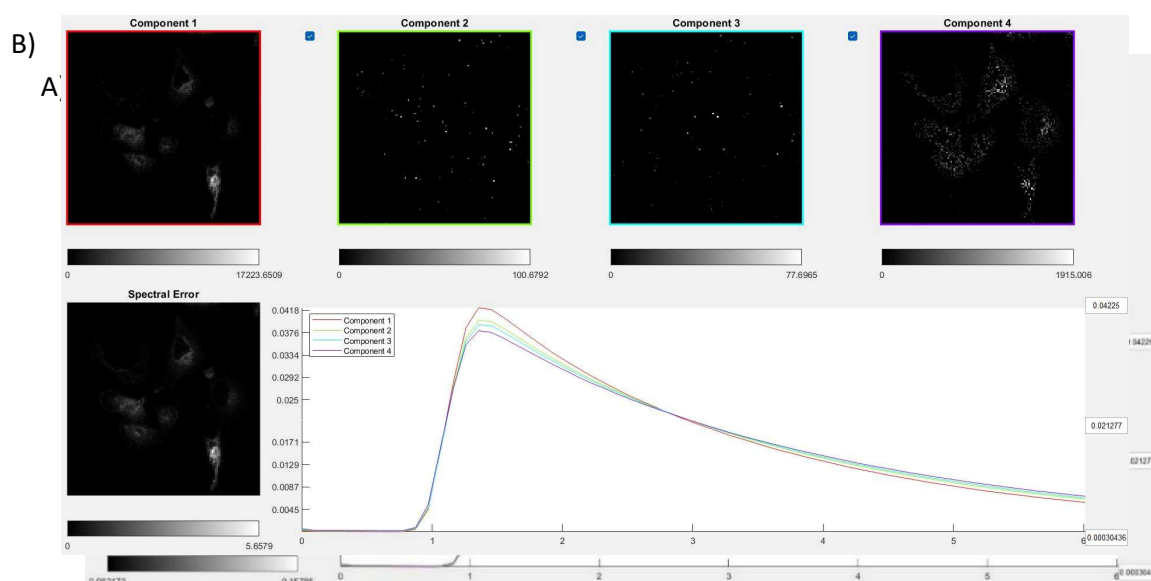


Figure 10. Image matrix showing four components corresponding to the spatial concentration maps. A) Result of pH 5.0 recombinant dataset showing component 1 as dominant B) Result of pH 6.5 recombinant dataset showing component 3 as dominant

The same guided NMF procedure was then applied to the cellular probe datasets. Probe data is the experimental FLIM dataset acquired from living cells expressing organelle targeted RpHLuorin2 constructs, used to measure and map the actual pH inside the organelles. The spectral range was limited to 6 ns, and binning was set to 4 x 4. Using FSC3, the number of components was set to 4 with minimum and maximum iterations set to 1. The single method was selected, guided analysis and partial noise whitening were enabled, and the ASCII reference file was provided. The results of the guided analysis on cellular data can be seen in figure 11, which shows the image matrix with four components, each corresponding to the spatial concentration maps for the four reference pH values.

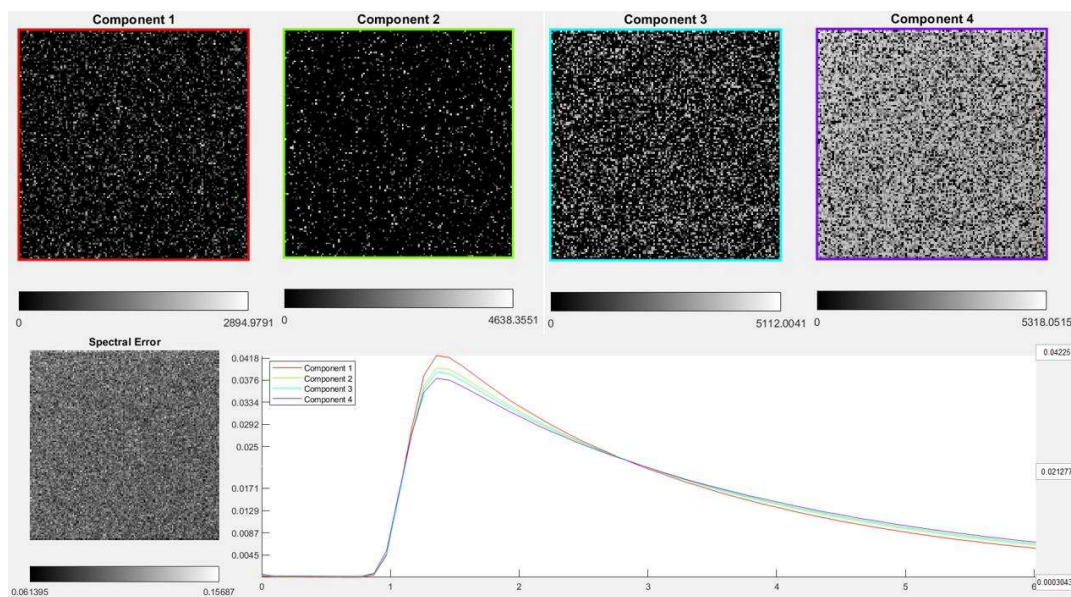


Figure 11. Image matrix showing the FSC3 analysis results of a pH 4.5 probe dataset. Component 1 can be sent as dominant

2.6 Calculation and Visualisation of Quantitative pH Maps

The concentration maps generated by NMF were further processed to generate quantitative pH maps using a weighted average approach. The saved resultant concentration maps, which indicate how much each pH component contributes to each pixel (as seen in figure 10), were imported into ImageJ software for quantitative synthesis (Jensen 2013). For each pixel, a weighted average pH was calculated using the formula: $pH_{estimated} = (\sum_i pH_i C_i) / (\sum_i C_i)$, where i is the component index, pH_i is the nominal pH of each reference

component (4.5, 5.5, 6.0, 7.5), and C_i is the abundance (concentration) of that component in the pixel.

To make visualisation of the pH values easier, a custom MATLAB script was employed. The script calculated the pH using the weighted average formula and mapped the resulting pH value to a hue value. The pH range of 4.5 – 7.5 was linearly mapped to a hue range of 0-1. The normalised sum of the concentration maps was used to determine the value or brightness and saturation, with saturation kept to 1 for all pixels. The resulting HSV (Hue, Saturation, and Value) image was then converted to RGB for display (Illustrated in figure 12). This colour-coded representation allows for easier visual assessment of pH distribution across the sample.

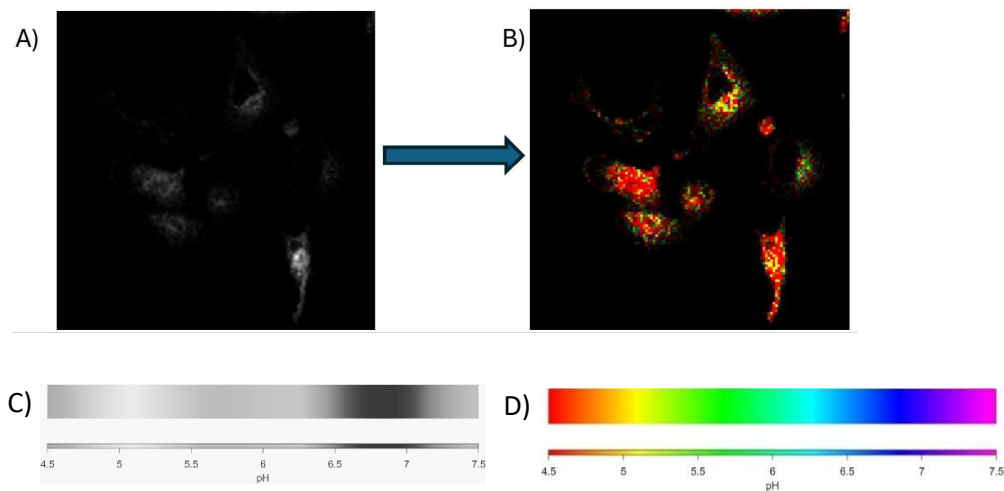


Figure 12. Images after spectral analysis showing the organelle. A pH4.5 probe dataset was used. A) Non-coloured image map B) Coloured image map showing a dominating red colour C) Grayscale bar D) pH colourbar

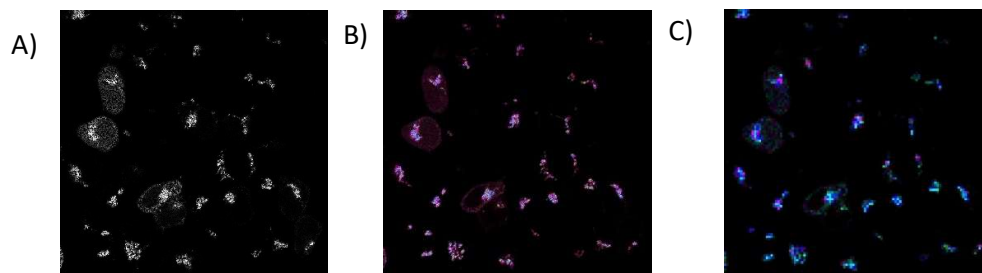
3. Results

3.1 Impact of Partial Noise Whitening on Data Quality

The effect of partial noise whitening on the quality of the factorisation was assessed by comparing results obtained with and without this pre-processing step. As illustrated in figure 8, analysis performed without partial noise whitening resulted in pH maps containing excessive noise and poor spatial consistency (Chen et al. 2023). This could be due to the algorithm being overwhelmed by the high variance in the early decay region, which caused it to fail in extracting meaningful component distribution from the data. When partial noise whitening was applied, the resulting pH maps showed improved clarity, with more coherent spatial structures and reduced pixel fluctuations. This confirms the importance of appropriate noise handling to obtain stable NMF solutions when working with photon counting FLIM data.

3.2 Effect of Spatial Binning on pH Map Quality

The importance of spatial binning to get a good quality pH map was studied. Figure 7 shows the comparison of results obtained with different binning factors. Analysis performed without binning (1 x 1) produced pH maps dominated by noise as seen in figure 13(A) and (B), with large variations between each pixel that rendered the data unreliable for consistently determining the pH values. As the binning factor was increased, image quality and signal to noise ratio (SNR) progressively improved. At 4 x 4 binning, the pH map showed clear spatial coherence with regions displaying uniform pH (Figure 13(C)). At higher binning factors of 8 x 8 and above, while the SNR continued to improve, spatial resolution was subsequently lost, with fine structural details becoming increasingly blurred. Based on this evaluation, 4 x 4 binning was selected as it offered an optimal compromise between SNR improvement and spatial resolution preservation for all subsequent analyses.



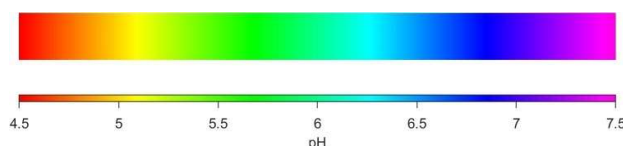


Figure 13. The images A and B show the results without any binning. Image C is the result with 4x4 binning which gives a better SNR map.

3.3 Validation of Guided Analysis Using Recombinant Calibrated Data

To ensure that the analysis framework performs as expected, without any systematic errors, the guided NMF method was first tested using recombinant RpHLuorin2 datasets of known pH. When analysis of a pH 7.5 recombinant sample was performed, the resulting pH map aligned with the expected outcome. After applying the RGB colour mapping, the image showed violet, the hue assigned to high pH in this colour scheme, as seen in figure 14(D). The colour coded map helped determine the approximate pH value through visual assessment across the image. Histogram analysis of this result showed a peak at approximately pH 7.3. The assignment of colours to pH values depends on the specific look-up table (LUT) employed. For this analysis, violet represented higher pH and red represented lower pH, and this mapping was consistently applied throughout all analyses.

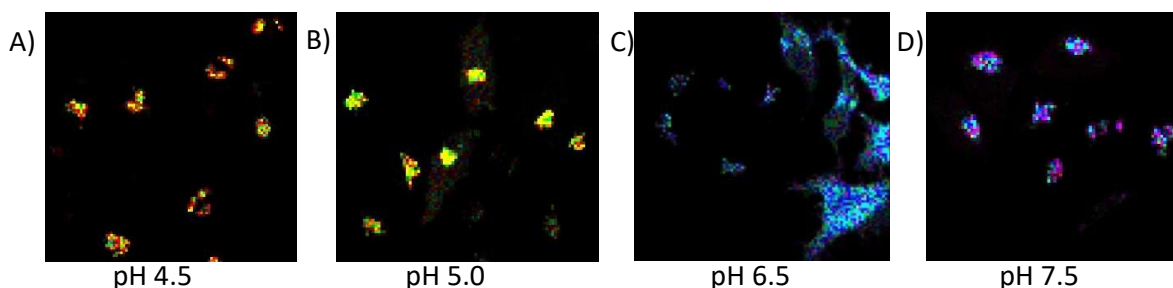


Figure 14. Colour-coded pH maps demonstrating that the algorithm can correctly assign pH values based on the reference decay. A) Predominantly red corresponding to pH 4.5 B) Predominantly yellow corresponding to pH 5.0 C) Predominantly light blue corresponding to pH 6.5 D) Predominantly violet corresponding to pH 7.5 E) pH colourbar

When a pH 4.5 recombinant sample was analysed, the resulting pH map was predominantly red as seen in figure 14 (A), the hue assigned to low pH, and the histogram peaked at pH 4.7. This demonstrates that the algorithm is capable of correctly assigning the correct pH values based on the reference decay curves. Furthermore, other intermediate pH

recombinant samples (5.0 and 6.5) were also analysed, each showing colour distributions consistent with their expected pH values, as illustrated in figure 14(B) and (C).

3.4 Application to Cellular Probe Data

After successful validation of the calibration data, the guided analysis method was applied to probe FLIM data acquired from living cells expressing organelle targeting RpHLuorin2 constructs. Each cellular probe dataset has an associated expected pH value based on the targeted organelle (e.g. ER ~ 7.2 , Golgi ~ 6.0 - 6.5). The MATLAB script was used to encode pH as a colour, generating colour coded pH maps for visual assessment of their values.

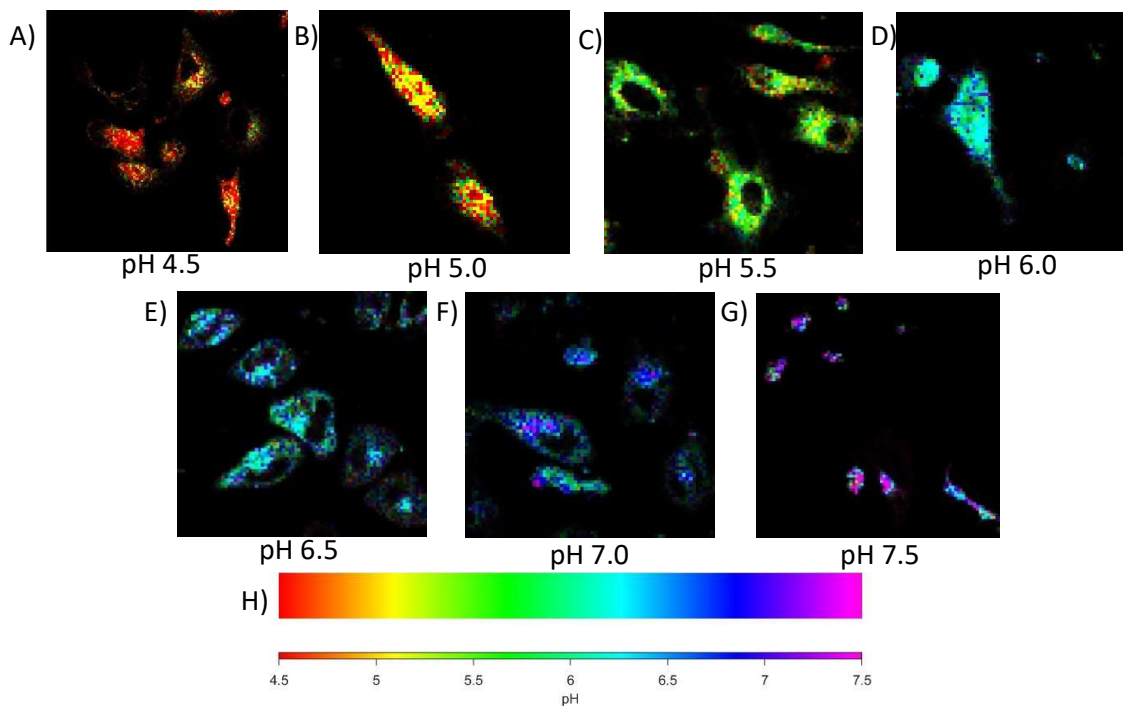


Figure 15. Colour-coded pH maps showing the different pH values. Each image corresponds with the pH when compared to the colourbar.

Figure 15 presents a selection of cellular probe datasets that showed good agreement between calculated and expected pH values. The colour distributions across these images were consistent with the anticipated pH ranges for their respective organelles, each showing coherent spatial structures. The sample that targeted ER in figure 15(G) displays

predominantly violet hues consistent with near-neutral pH, while those that targeted Golgi in figure 15(D) and (E) show bluish-green colours corresponding to their moderately acidic environment. The lysosomal sample in figure 15(A) exhibits red hues consistent with highly acidic conditions. These results successfully show that the guided NMF method can be used to generate interpretable pH maps from cellular FLIM data when experimental conditions are optimal.

However, some cellular probe datasets produced results that deviated from the expected values, as seen in figure 16. In these cases, the colour coded pH maps showed results inconsistent with the expected pH of the organelle. Moreover, some images exhibited poor spatial structure with increased fragmentation. It is important to note that the analysis procedure applied to these datasets was identical to that used for the successful examples in figure 15. They had the same binning factor (4x4), reference curves, and guided NMF parameters. Such discrepancies therefore cannot be attributed to the analytical method itself.

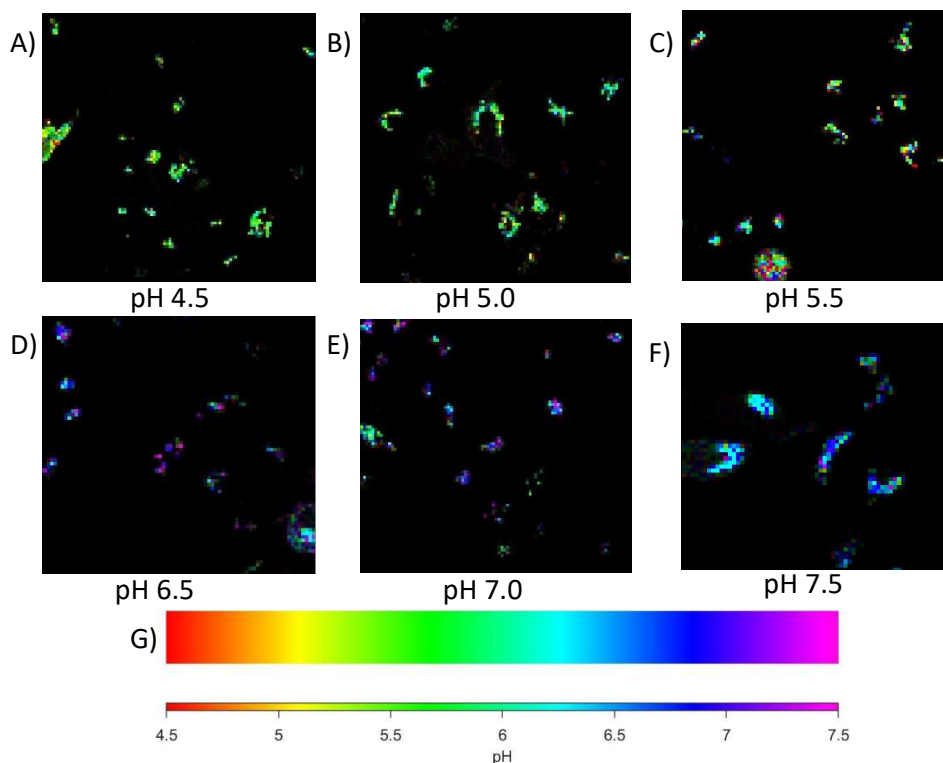


Figure 16. Probe dataset analysis results exhibiting poor spatial structure and inaccurate colour corresponding to the expected pH value

A possible explanation for such variabilities is that the datasets shown in figure 15 and figure 16 were acquired on different experimental days. The variations caused by the different experimental conditions on different days could affect data quality. Other explanations could be lower excitation power resulting in reduced photon counts and poorer SNR, incomplete pH stabilization of the cellular environment, or slight misalignment of the optical system, all of which could contribute to the degraded data quality. Without detailed information on the conditions under which each dataset was acquired, it is extremely difficult to identify the cause of the variability.

However, since some datasets produced excellent results while others did not, despite identical analysis procedures, it could suggest that such variable results are mainly due to the raw data quality rather than the analytical method. This highlights the importance of consistent experimental conditions and sufficient photon budget for reliable FLIM pH mapping (Nasser and Meller 2021).

4. Discussion

4.1 Integration of Experimental Findings with Existing Literature

The results of this project both validate and critically qualify principles from existing literature on FLIM data analysis. This method proved successful in differentiating between a broad range of pH values and was particularly effective at classifying samples as acidic or neutral. The validation on recombinant calibration data confirmed that the algorithm correctly assigned pH values based on the reference decay curves across the full pH range of 4.5 – 7.5. This confirms the validity of the foundational principle that the lifetime of RpHLuorin2 is a reliable proxy for pH and that multi-component analysis can yield good spatial image results for different pH states. This finding aligns with the work of Linders et al. (2022).

The successful generation of interpretable pH maps from cellular probe datasets (Figure 15) demonstrates that the guided NMF approach can be applied to complex biological samples. The ER targeted sample displayed violet hues consistent with near neutral pH, while Golgi targeted samples showed intermediate colours corresponding to their moderately acidic environment, and the lysosomal sample exhibited red hues consistent with highly acidic conditions. These results validate the practical usage of the method for organelle pH mapping when experimental conditions are optimal.

However, the systematic observation that some cellular datasets produced poor results despite identical procedures (figure 16) means that caution must be exercised. The photophysical behaviour of a fluorophore *in cellulo* can yield different results from those obtained through *in vitro* calibration due to factors such as macromolecular crowding, viscosity, and specific local interactions. Using the supervised approach, which employed *in vitro* references as guides during analysis, made this method vulnerable to such divergence when reference dynamics imperfectly matched cellular conditions. This finding directly aligns with the warning by Masia et al. (2022) regarding the risks associated with using fixed reference patterns when environmental conditions between reference and experimental samples are not perfectly matched.

The variability between successful and unsuccessful cellular datasets cannot be attributed to the analytical method itself, as the same procedure was applied to every analysis. The

datasets shown in figure 15 and figure 16 were acquired on different experimental days, suggesting that even minor variations in experimental conditions can affect data quality. Factors such as lower excitation power resulting in reduced photon counts and poorer SNR, incomplete pH stabilization of the cellular environment, or slight optical misalignment could all contribute to reduced data quality. This highlights the importance of consistent experimental conditions and sufficient photon budget for reliable FLIM pH mapping.

4.2 The Photon Budget and Its Implications

The necessity for spatial binning highlights one of the major constraints in FLIM, which is the photon budget. As demonstrated in figure 7, analysis performed without binning (1x1) produced pH maps dominated by noise, with large variations seen in each pixel that rendered the data unreliable for quantitative interpretation. Only with 4 x 4 binning did the pH maps show clear spatial structures with well-defined regions of uniform pH.

The precision of any parameters extracted from photon-shot-noise-limited data scales with $1/\sqrt{N}$, where N is the photon count. This means that the requirement to bin data to obtain meaningful results directly sacrifices spatial resolution to achieve a reliable lifetime reading or SNR per analysed unit (Callenberg et al. 2021). Such a trade-off must be carefully considered when designing live-cell FLIM experimental analysis. The optimal binning factor of 4 x 4 represents the best compromise between SNR improvement and spatial resolution preservation for this experimental analysis.

Noise handling is an important part of data analysis and it was further emphasised by the results after using partial noise whitening (figure 8). Analysis performed without whitening resulted in pH maps containing excessive noise and poor spatial coherence, as the algorithm was overwhelmed by the high variance in the early decay region. When partial noise whitening was applied, the resulting pH maps showed an improved clarity. This confirms that testing for the best conditions and settings is essential for NMF solutions when working with photon counting FLIM data. This is consistent with the approach used to validate the uFLIM algorithm (Masia et al. 2022).

4.3 Background Noise: An Ongoing Issue

From the pH image maps, noise was consistently observed as a factor affecting data quality. This occurs when objects are close to each other, overlapping, or touching during spectral imaging (Amigo et al. 2015). Moreover, organelles exhibit different spectral profiles, which directly affects the spectra of the background adjacent to the object being imaged, potentially causing them to share similar spectral characteristics. This makes the removal of noise challenging with automated methods, as differentiating between signal and background becomes difficult.

To mitigate the noise issue, an identification mark could be assigned to each pixel belonging to the structure under analysis, classifying them into the same class. Any other pixels would then be removed, leaving an image free of background noise. This method proposed by Amigo et al. (2015), however, would be extremely time consuming and highly dependent on correctly identifying the organelle and its boundaries, introducing potential for user bias. Since our analysis experiment was local, meaning the results on a pixel do not affect the results on other pixels, we would only experience an issue in overlapping regions and not in the pixels within the same organelle. This independence of pixel-wide analysis can be both a strength, to help avoid error propagation, and a limitation due to the inability to use spatial context for noise reduction.

4.4 The Conceptual Framework of NMF and its Application

The NMF algorithm used in this analysis provides an interpretable framework for unmixing complex FLIM data. It enables the separation of signals measured from the decay curves at each pixel from different sources. In the guided NMF approach, the temporal dynamics matrix T was fixed using the pre-generated reference decay curves corresponding to pH 4.5, 5.5, 6.0, and 7.5. This approach uses the pure component spectra as a guide to improve the stability and interpretability of the decomposition. The algorithm solves only for the spatial concentration matrix S to determine how much of each reference pH component contributes to every pixel in the image. The resulting concentration maps, as seen in figures 10 and 11, help visualise the spatial distribution of each pH state. Calculating the weighted average then converts the four component maps into a single quantitative pH map.

It should be noted that an initial analysis attempt was made using the unsupervised method, where a reference spectra was not provided and both S and T were allowed to change freely. However, this approach did not result in favorable results on analysing the data, likely due to the low photon budget and the subtle differences between closely spaced pH states. This discovery led to the adoption of the guided approach, where a reference component spectra was used to constrain the solution.

4.5 The Pathway Forward: Moving from Supervised Guidance to Unsupervised Discovery

The limitations identified in this study can be used to suggest the development of a new analytical framework, representing a logical progression from our supervised, model-constrained approach to unsupervised, data driven approach such as uFLIM (Masia et al. 2022) and phasor S-FLIM (Digman et al. 2007) (Scipioni et al. 2021). Implementing the uFLIM algorithm on cellular FLIM data could address the issues by directly addressing the rigidity of the analysis model. Unlike guided NMF, which fixes component dynamics to *in vitro* references, uFLIM determines both spatial distribution and temporal dynamics simultaneously from the data being analysed, without requiring pre-defined reference spectra.

This unsupervised approach could also test the decay of organelle targeting RpHLuorin2 without bias, potentially revealing intracellular environmental heterogeneity within an organelle that could not be confirmed by the discrete four component model. Due to its ability to isolate autofluorescence as a separate component using the distinct lifetime-spectrum profiles of endogenous fluorophores, uFLIM could provide a clearer pH signal by separating background noise from specific signals (Hwang et al. 2025). This would be particularly useful for cellular imaging where autofluorescence from NADH, FAD, and other endogenous fluorophores can contaminate the pH sensor signal.

Employing a hybrid initialisation strategy by combining the current model with the proposed approach could improve data robustness (Kervrann et al. 2015). The guided analysis method, using the two extreme pH endpoints (4.5 and 7.4), could provide an initial estimate for the spatial distribution of protonated and deprotonated sensor populations. Using these estimates as an initialisation or starting point, unsupervised FLIM or phasor S-FLIM analysis could then be applied. This would accelerate convergence, ensuring the result closely

approximates the actual pH value while allowing the algorithm the freedom to discover the nature of all components present in the cellular data, including potential intermediate states or unexpected contributions.

The ultimate goal of the analysis is to obtain clear, undistorted, multiplexed, and dynamic image readouts. To achieve this, we could co-express a FLIM pH sensor (RpHLuorin) with a FRET-based biosensor for calcium, cAMP, or kinase activity (Verma et al. 2023). This would extend the analysis beyond simply calculating pH values and move towards a multi-parameter cellular physiology. Independent pH maps and biochemical activity within a living cell could be computed simultaneously using a single excitation wavelength alongside μ FLIM-FRET or phasor S-FLIM. The result would provide insights into coupled physiological processes, such as the correlation between Golgi pH levels and calcium signalling during secretion.

4.6 Conclusion

This project has successfully implemented a guided unmixing FLIM methodology for intracellular pH mapping. Following rigorous evaluation, spatially coherent pH maps were consistently generated that distinguish broad environmental states, validating the key principles of lifetime-based pH sensing. The validation on recombinant calibration data confirmed accurate pH assignment across the full range of 4.5-7.5. This was successfully applied to cellular probe datasets which demonstrated the practical utility of the method for mapping organelle pH.

Through critical evaluation, we have identified the method's major drawbacks, which included its susceptibility to systematic error when reference dynamics do not match cellular conditions, its compromised resolution due to SNR limits, and its inherent model rigidity when constrained to four components. The differences in results observed between some cellular datasets, despite identical analysis procedures, highlights the importance of consistent experimental conditions and sufficient photon budget for reliable results. The photon budget constraint resulted in the necessity for 4x4 spatial binning, representing the trade-off between SNR and spatial resolution that must be carefully managed.

These findings provide justification for evolving the methodology in FLIM data analysis. Integrating robust experimental methodologies using quantitative sensors with the superior

analytical power of unsupervised computational framework, such as uFLIM, could lead to more refined quantitative subcellular pH mapping. By seamlessly combining data, methods, and theories to create a cohesive analysis approach, we can move beyond qualitative maps to achieve quantitative, dynamic, and multi-parametric image maps of cellular physiology. This would help unlock deeper analytical insights in both health and disease, contributing to our understanding of fundamental cell biology and potentially discovering strategies for combating diseases where pH homeostasis is disrupted.

References

- Adam, E., Mutanga, O. and Rugege, D. 2009. Multispectral and hyperspectral remote sensing for identification and mapping of wetland vegetation: a review. *Wetlands Ecology and Management* 18(3), pp. 281–296. Available at: <https://doi.org/10.1007/s11273-009-9169-z>.
- Amigo, J.M., Babamoradi, H. and Elcoroaristizabal, S. 2015. Hyperspectral image analysis. A tutorial. *Analytica Chimica Acta* 896, pp. 34–51. Available at: <https://doi.org/10.1016/j.aca.2015.09.030>.
- Amigo, J.M., Martí, I. and Gowen, A. 2013. Hyperspectral imaging and chemometrics. In: *Data handling in science and technology*. pp. 343–370. Available at: <https://doi.org/10.1016/b978-0-444-59528-7.00009-0>.
- Anderson, P.E., Mahle, D.A., Doom, T.E., Reo, N.V., DelRaso, N.J. and Raymer, M.L. 2010. Dynamic adaptive binning: an improved quantification technique for NMR spectroscopic data. *Metabolomics* 7(2), pp. 179–190. Available at: <https://doi.org/10.1007/s11306-010-0242-7>.
- Berezin, M.Y. and Achilefu, S. 2010. Fluorescence lifetime measurements and biological imaging. *Chemical Reviews* 110(5), pp. 2641–2684. Available at: <https://doi.org/10.1021/cr900343z>.
- Boorman. 2019. *Multimodal Coherent Anti-Stokes Raman Scattering (CARS) Microscopy For The Detection Of Isotope-Labelled Molecules*. Cardiff University School of Biosciences. Available at: <https://orca.cardiff.ac.uk/id/eprint/130291/1/DaleBoormanCorrectedThesisFinalVersion.pdf> [Accessed: 11 February 2026].
- Bower, A.J., Li, J., Chaney, E.J., Marjanovic, M., Spillman, D.R. and Boppart, S.A. 2018. High-speed imaging of transient metabolic dynamics using two-photon fluorescence lifetime imaging microscopy. *Optica* 5(10), p. 1290. Available at: <https://doi.org/10.1364/optica.5.001290>.
- Callenberg, C. et al. 2021. Super-resolution time-resolved imaging using computational sensor fusion. *Scientific Reports* 11(1), p. 1689. Available at: <https://doi.org/10.1038/s41598-021-81159-x>.

- Chen, X. et al. 2023. Boosting quantification accuracy of chemical exchange saturation transfer MRI with a spatial–spectral redundancy-based denoising method. *NMR in Biomedicine* 37(1), p. e5027. Available at: <https://doi.org/10.1002/nbm.5027>.
- Coelho, L.P., Peng, T. and Murphy, R.F. 2010. Quantifying the distribution of probes between subcellular locations using unsupervised pattern unmixing. *Bioinformatics* 26(12), pp. i7–i12. Available at: <https://doi.org/10.1093/bioinformatics/btq220>.
- Datta, R., Heaster, T.M., Sharick, J.T., Gillette, A.A. and Skala, M.C. 2020. Fluorescence lifetime imaging microscopy: fundamentals and advances in instrumentation, analysis, and applications. *Journal of Biomedical Optics* 25(07), p. 1. Available at: <https://doi.org/10.1117/1.jbo.25.7.071203>.
- Digman, M.A., Caiolfa, V.R., Zama, M. and Gratton, E. 2007. The Phasor Approach to Fluorescence Lifetime Imaging Analysis. *Biophysical Journal* 94(2), pp. L14–L16. Available at: <https://doi.org/10.1529/biophysj.107.120154>.
- Gabaldón, T. and Pittis, A.A. 2015. Origin and evolution of metabolic sub-cellular compartmentalization in eukaryotes. *Biochimie* 119, pp. 262–268. Available at: <https://doi.org/10.1016/j.biochi.2015.03.021>.
- Gowen, A.A. and Amigo, J.M. 2013. Applications of Spectroscopy and Chemical Imaging in Pharmaceuticals. *Handbook for Biophotonics*, pp. 71–88. Available at: <https://doi.org/10.1002/9783527643981.bphot083>.
- Gregor, I. and Patting, M. 2014. Pattern-Based linear unmixing for efficient and reliable analysis of multicomponent TCSPC data. In: *Springer series on fluorescence*. pp. 241–263. Available at: https://doi.org/10.1007/4243_2014_70.
- Guo, J., Mai, F.-Y., Li, X.-Y., Liang, J.-R., Yang, W.-T. and Li, C.-G. 2025. Establishment of Rphluorin2-expressing cell and its application in monitoring JTC-801-induced alkaliptosis via multi-dimensional fluorescence detection approaches. *Frontiers in Cell and Developmental Biology* 13, p. 1727740. Available at: <https://doi.org/10.3389/fcell.2025.1727740>.
- Hwang, W., McPartland, T., Jeong, S. and Evans, C.L. 2025. A robust method for autofluorescence-free immunofluorescence using high-speed fluorescence lifetime imaging microscopy. *Scientific Reports* 15(1), p. 5503. Available at: <https://doi.org/10.1038/s41598-025-89142-6>.

- Jain, A., Blum, C. and Subramaniam, V. 2009. Fluorescence Lifetime spectroscopy and imaging of visible fluorescent proteins. In: *Elsevier eBooks*. pp. 147–176. Available at: <https://doi.org/10.1016/b978-0-444-53075-2.00004-6>.
- Jain, A. and Zoncu, R. 2022. Organelle transporters and inter-organelle communication as drivers of metabolic regulation and cellular homeostasis. *Molecular Metabolism* 60, p. 101481. Available at: <https://doi.org/10.1016/j.molmet.2022.101481>.
- Jensen, E.C. 2013. Quantitative analysis of histological staining and fluorescence using ImageJ. *The Anatomical Record* 296(3), pp. 378–381. Available at: <https://doi.org/10.1002/ar.22641>.
- Kamiya, Y., Kamiya, D., Yamamoto, K., Nyfeler, B., Hauri, H.-P. and Kato, K. 2007. Molecular basis of sugar recognition by the human L-type lectins ERGIC-53, VIPL, and VIP36. *Journal of Biological Chemistry* 283(4), pp. 1857–1861. Available at: <https://doi.org/10.1074/jbc.m709384200>.
- Kervrann, C., Sorzano, C.O.S., Acton, S.T., Olivo-Marin, J.-C. and Unser, M. 2015. A guided tour of selected image processing and analysis methods for fluorescence and electron microscopy. *IEEE Journal of Selected Topics in Signal Processing* 10(1), pp. 6–30. Available at: <https://doi.org/10.1109/jstsp.2015.2505402>.
- Khan, M.J., Khan, H.S., Yousaf, A., Khurshid, K. and Abbas, A. 2018. Modern Trends in Hyperspectral Image Analysis: A review. *IEEE Access* 6, pp. 14118–14129. Available at: <https://doi.org/10.1109/access.2018.2812999>.
- Lee, C.A., Gasster, S.D., Plaza, A., Chang, C.-I. and Huang, B. 2011. Recent developments in High performance Computing for Remote sensing: a review. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 4(3), pp. 508–527. Available at: <https://doi.org/10.1109/jstars.2011.2162643>.
- Linders, P.T.A., Ioannidis, M., Ter Beest, M. and Van Den Bogaart, G. 2022. Fluorescence Lifetime Imaging of pH along the Secretory Pathway. *ACS Chemical Biology* 17(1), pp. 240–251. Available at: <https://doi.org/10.1021/acscchembio.1c00907>.
- Linders, P.T.A., Peters, E., Ter Beest, M., Lefeber, D.J. and Van Den Bogaart, G. 2020. Sugary logistics gone wrong: membrane trafficking and congenital disorders of glycosylation. *International Journal of Molecular Sciences* 21(13), p. 4654. Available at: <https://doi.org/10.3390/ijms21134654>.

- Liu, Y., Kim, H.-R. and Heikal, A.A. 2006. Structural Basis of Fluorescence Fluctuation Dynamics of Green Fluorescent Proteins in Acidic Environments. *The Journal of Physical Chemistry B* 110(47), pp. 24138–24146. Available at: <https://doi.org/10.1021/jp062164t>.
- Mahon, M.J. 2011. pHluorin2: an enhanced, ratiometric, pH-sensitive green fluorescent protein. *Advances in Bioscience and Biotechnology* 02(03), pp. 132–137. Available at: <https://doi.org/10.4236/abb.2011.23021>.
- Martinez, K.A., Kitko, R.D., Mershon, J.P., Adcox, H.E., Malek, K.A., Berkmen, M.B. and Slonczewski, J.L. 2012. Cytoplasmic pH Response to Acid Stress in Individual Cells of *Escherichia coli* and *Bacillus subtilis* Observed by Fluorescence Ratio Imaging Microscopy. *Applied and Environmental Microbiology* 78(10), pp. 3706–3714. Available at: <https://doi.org/10.1128/aem.00354-12>.
- Martinez, N.J., Titus, S.A., Wagner, A.K. and Simeonov, A. 2015. High-throughput fluorescence imaging approaches for drug discovery using in vitro and in vivo three-dimensional models. *Expert Opinion on Drug Discovery* 10(12), pp. 1347–1361. Available at: <https://doi.org/10.1517/17460441.2015.1091814>.
- Masia, F., Dewitte, W., Borri, P. and Langbein, W. 2022. uFLIM — Unsupervised analysis of FLIM-FRET microscopy data. *Medical Image Analysis* 82, p. 102579. Available at: <https://doi.org/10.1016/j.media.2022.102579>.
- Masia, F., Glen, A., Stephens, P., Borri, P. and Langbein, W. 2013. Quantitative chemical imaging and unsupervised analysis using hyperspectral coherent Anti-Stokes Raman Scattering microscopy. *Analytical Chemistry* 85(22), pp. 10820–10828. Available at: <https://doi.org/10.1021/ac402303g>.
- MATLAB. [no date]. *Get Started with Hyperspectral Image Processing*. Available at: <https://www.mathworks.com/help/images/get-started-with-hyperspectral-image-processing.html> [Accessed: 11 February 2026].
- Nasser, M. and Meller, A. 2021. Lifetime-based analysis of binary fluorophores mixtures in the low photon count limit. *iScience* 25(1), p. 103554. Available at: <https://doi.org/10.1016/j.isci.2021.103554>.
- Ng, T.L.C., Capalbo, L. and Kumita, J.R. 2025. Microscopy-based techniques for studying the material properties of biomolecular condensates in the cellular environment. *Biophysical Reviews* 17(4), pp. 1119–1132. Available at: <https://doi.org/10.1007/s12551-025-01343-5>.

- Petukh, M., Stefl, S. and Alexov, E. 2013. The role of protonation states in Ligand-Receptor recognition and binding. *Current Pharmaceutical Design* 19(23), pp. 4182–4190. Available at: <https://doi.org/10.2174/1381612811319230004>.
- Rennick, J.J., Nowell, C.J., Pouton, C.W. and Johnston, A.P.R. 2022. Resolving subcellular pH with a quantitative fluorescent lifetime biosensor. *Nature Communications* 13(1), p. 6023. Available at: <https://doi.org/10.1038/s41467-022-33348-z>.
- Royle, S.J., Granseth, B., Odermatt, B., Derevier, A. and Lagnado, L. 2008. Imaging PHluorin-Based probes at hippocampal synapses. *Methods in Molecular Biology* 457, pp. 293–303. Available at: https://doi.org/10.1007/978-1-59745-261-8_22.
- Salis, A. and Monduzzi, M. 2016. Not only pH. Specific buffer effects in biological systems. *Current Opinion in Colloid & Interface Science* 23, pp. 1–9. Available at: <https://doi.org/10.1016/j.cocis.2016.04.004>.
- Schulte, A., Lorenzen, I., Böttcher, M. and Plieth, C. 2006. A novel fluorescent pH probe for expression in plants. *Plant Methods* 2(1), p. 7. Available at: <https://doi.org/10.1186/1746-4811-2-7>.
- Scipioni, L., Rossetta, A., Tedeschi, G. and Gratton, E. 2021. Phasor S-FLIM: a new paradigm for fast and robust spectral fluorescence lifetime imaging. *Nature Methods* 18(5), pp. 542–550. Available at: <https://doi.org/10.1038/s41592-021-01108-4>.
- Soltani, S., Paulson, J.G., Fong, E.J., Mumenthaler, S.M. and Armani, A.M. 2025. Denoising of fluorescence lifetime imaging data via principal component analysis. *Scientific Reports* 15(1), p. 40258. Available at: <https://doi.org/10.1038/s41598-025-24022-7>.
- Torrado, B., Pannunzio, B., Malacrida, L. and Digman, M.A. 2024. Fluorescence lifetime imaging microscopy. *Nature Reviews Methods Primers* 4(1). Available at: <https://doi.org/10.1038/s43586-024-00358-8>.
- Verma, A.K., Noumani, A., Yadav, A.K. and Solanki, P.R. 2023. FRET based biosensor: principle applications recent advances and challenges. *Diagnostics* 13(8), p. 1375. Available at: <https://doi.org/10.3390/diagnostics13081375>.
- Warren, S.C. et al. 2013. Rapid global fitting of large fluorescence lifetime imaging microscopy datasets. *PLoS ONE* 8(8), p. e70687. Available at: <https://doi.org/10.1371/journal.pone.0070687>.
- Wu, D. and Minton, A.P. 2015. Quantitative characterization of nonspecific self- and Hetero-Interactions of proteins in nonideal solutions via static light scattering. *The Journal of*

Physical Chemistry B 119(5), pp. 1891–1898. Available at:

<https://doi.org/10.1021/jp510918d>.

Wüstner, D. 2022. Dynamic mode decomposition of fluorescence loss in photobleaching microscopy data for Model-Free analysis of protein transport and aggregation in living cells.

Sensors 22(13), p. 4731. Available at: <https://doi.org/10.3390/s22134731>.

Zhang, C., Wang, H., Huang, J. and Gao, Q. 2014. The visible to the near infrared narrow band acousto-optic tunable filter and the hyperspectral microscopic imaging on biomedicine study. *Journal of Optics* 16(12), p. 125303. Available at: [https://doi.org/10.1088/2040-](https://doi.org/10.1088/2040-8978/16/12/125303)

[8978/16/12/125303](https://doi.org/10.1088/2040-8978/16/12/125303).

Zhang, X., Zhang, X., Cheng, S., Fan, X., Bao, H., Zhou, S. and Ping, J. 2024. Spatiotemporal cell control via High-Precision electronic regulation of microenvironmental PH. *Nano Letters* 24(49), pp. 15645–15651. Available at: <https://doi.org/10.1021/acs.nanolett.4c04174>.